

John Balci

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EDUCATION

Purdue University

West Lafayette, IN

B.S. in Electrical Engineering, Conc. Software Engineering | GPA: 3.58/4.0

Aug. 2025 – May 2028 (Expected)

EXPERIENCE

TRMC Intern – Telemetry Software and Instrumentation

Summer 2026

Naval Test Wing Atlantic, Advanced Systems Group (Hired by ASRC Federal TRMCi)

Patuxent River, MD

- Architected an automated IRIG-106 telemetry frame synthesizer in Python using a **combinatorial solver** to optimize 2D PCM grid dimensions, balancing RF bandwidth, frame overhead, and sample rates.
- Engineered a **2D Backtracking CSP solver** with forward checking and **6-metric heuristic** to place sensor data into PCM maps; integrated automated color-coded Excel exports, automating **20-40 hour process** to seconds.
- Supported E-2D Hawkeye flight test operations across a 3-aircraft fleet for VX-20 Squadron.
- Resolved a critical GPS antenna fault prior to flight, restoring satellite telemetry and preventing delays.

Undergraduate Researcher – Avionics & Embedded Hardware

Aug. 2025 – Present

Purdue Cubesat Lab, Electric Propulsion and Plasma Lab

West Lafayette, IN

- Synthesized 1V half-wave sinusoidal pulses (20 microsecond duration, 1Hz PRF) in **MATLAB** to characterize discharge cycles for a novel **pulsed-magnetoplasma dynamic** (PMPD) thruster in a 6U CubeSat.
- **Engineered a precision capacitive gauging system** for propellant mass measurement. Utilized CAPDAC to offset parasitic capacitance at a high-resolution 15pF dynamic range.
- **Developed fringing field correction factors** for parallel-plate capacitance model to account for plate separation-to-size ratios, ensuring accuracy in 0.23 mm displacement measurements.

Founder, Principal Owner, Scrappr.site

Sept 2025 – Present

JEB Digital Ventures

Fairfax, VA

- Architected a **semantic search engine** using OpenAI text embeddings and weighted cosine similarity math to rank faculty profiles against research queries using OpenAlex API publication data **with 26 users**.
- Developed an **end-to-end AI outreach pipeline** transforming candidate CVs and relevant academic metadata into personalized emails and tailored cover letters **based on researcher's recent outputs**.

AI Integration Intern

Summer 2025

ChainwiseCPA

Reston, VA

- Built 3 scalable **API endpoint integrations** in JavaScript connecting enterprise accounting platforms, automating workflows supporting over \$1.7M in client transactions and saving hours per client deal.
- Designed and implemented 4 real-time AI project management automations using Zapier custom workflows and Claude API, improving operational efficiency.

Lead Propulsion Engineer

2021 – 2025

Caelus Rocketry

Alexandria, VA

- Led a 17-student liquid propulsion team & avionics system to the complete design, manufacturing, and integration of a liquid-fueled rocket engine, culminating in the successful **completion of cold flow tests and full readiness** for static fire.

PROJECTS

Gyroscopic Stabilized Pen | C++, OpenCV, ESP32-S3, Raspberry Pi 5, Git

2024 – 2025

- Built an assistive writing device for users with essential tremor and cerebral palsy, programming a closed-loop computer-vision and microcontroller stabilization system utilizing sensor fusion for accurate detection.

TECHNICAL SKILLS

EE: KiCad, LTSpice, MATLAB/Simulink, Circuit Analysis & Design, PCBs, MOSFETs, Verilog

Embedded: STM32, ESP32, Arduino, Raspberry Pi, Embedded C/C++, UART/SPI/I2C, Sensor Interfacing, Jetson

Instrumentation & Test: Telemetry Systems, Capacitive Measurement, Signal Synthesis, Data Acquisition

MCAD: Fusion 360, SolidWorks, Onshape, Autodesk Inventor, CNC Machining

Coding: Java, Python, C/C++, Swift, SQL (Postgres), JavaScript/Typescript, HTML/CSS, MATLAB